Bryce Grant

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EDUCATION & HONORS

Case Western Reserve University *May 2029 (Expected)*
PhD Candidate Robotics Graduate Advisor: Peng (Edward) Wang

- **Expertise:** Computer Vision, LLM Evals and Benchmarks, Mechanistic Interpretability, Multimodal Learning, RL, Robotics
- **Award:** ***NSF Graduate Research Fellow (GRFP 2025), Advanced to Candidacy, NVIDIA Academic Grant***
- **Leadership:** MATLAB Ambassador, NSBE Region III Finance Chair, Undergraduate Mentorship
- **Reviewer:** ICRA, ICLR, CoRL, ECCV, IROS
- **Invited Talk:** Evolution of Robotic Foundation Models, Free University of Bozen-Bolzano (Oct 2026)

University of Kentucky: *Magna Cum Laude, Lexmark Scholar* May 2024
Dual B.S. in Computer Engineering and Electrical Engineering (CS, Math Minors)

PUBLICATIONS & PREPRINTS

B. Grant*, X. Zhao*, P. Wang. Not All Features Are Created Equal: A Mechanistic Study of Vision-Language-Action Models.
NeurIPS 2026 | ICLR MM Intelligence Workshop (Oral), 2026 | Website
B. Grant and P. Wang. Gluing Local Contexts into Global Meaning: A Sheaf-Theoretic Decomposition of Transformer Representations.
NeurIPS 2026 | ICLR UCRL Workshop, 2026 | Website
B. Grant, A. Rothenberg*, L. Senning*, P. Wang. SPARK: Sequential Planning via Anchored Robotic Keypoints.
Under Review | Website
B. Grant, A. Rothenberg, A. Banerjee, P. Wang. TrianguLang: Geometry-Aware Semantic Consensus for Pose-Free 3D Localization.
Preprint | Website
B. Grant and P. Wang. Quaternion Approximate Networks for Enhanced Image Classification and Oriented Object Detection.
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS Oral), 2025 | Website

EXPERIENCE

Case Western Reserve University Cleveland, OH
Graduate Research Assistant Aug. 2024 – Present

- Co-led the first large-scale mechanistic interpretability study of VLA models, released a ∼7TB dataset and maintain Action Atlas
- Built SPARK, a training-free neurosymbolic agentic framework for manipulation with closed-loop verify-and-recover: best zero-shot result on LIBERO-Pro (75.1%), deployed physically on a UR10e, a Franka FR3 and a bimanual rig (68% over 220 physical trials)
- Researching WAM interpretability through probing and causal analysis of whether the action reads its own imagined future (blog)
- Developing a timed-execution testbed in MuJoCo-Warp with Isaac Sim cross-checks, audio and tactile verification, and ORCA dexterous-hand grasps, extending SPARK's verify-and-recover loop to guitar playing under hard timing deadlines

Mercor Remote
Applied AI Engineer Intern Aug. 2025 – Aug. 2026

- Full-stack engineering and delivery management for RL-environment and synthetic-data programs for coding agents across multi-million-dollar RLHF/SFT contracts: built environments and tooling, ran loss analyses, and shipped trajectory-annotation tools
- Built internal analytics pipelines (PySpark, Snowflake) tracking contractor throughput, task accuracy, and budget burn across projects

HP Vancouver, WA
PhD ML Intern May 2024 – Aug. 2024

- Deployed an anomaly detection pipeline on AWS for enterprise cloud spend analysis
- Engineered a comparative RAG framework: benchmarked semantic vs. recursive chunking, BM25 vs. FAISS retrieval, and LLM query expansion strategies, with evaluation via coherence and NDCG metrics
- Built a customer-facing LLM chat agent that guided users through printer setup, replacing static manuals with dynamic reranking and multi-stage error recovery

University of Kentucky Lexington, KY
AI for Smart Manufacturing Lab Research Aug. 2023 – May 2024

- Developed CNN-Transformer hybrid for 6-DOF pose estimation from RGB-D inputs for robotic arm manipulation
- Led navigation subsystem for autonomous telepresence robot, implementing 3D SLAM with servo-mounted LiDAR, achieving real-time mapping and dynamic obstacle avoidance in dynamic environments

Honeywell Atlanta, GA
Embedded Software Engineer Intern May 2023 – Aug. 2023

- Developed bare-metal firmware API for life safety microcontrollers, supporting product line with 100K+ unit volume

SKILLS

Programming Languages: C/C++, CUDA, MATLAB, Python, SQL
ML & Robotics: COLMAP, Distributed Training, HuggingFace, Isaac Sim / Lab, JAX, LiDAR, MuJoCo, OpenCV, Open3D, PyTorch, RL, ROS2, WandB, VLAs, VAMs/WAMs
Cloud, DevOps & Engineering: Blender, Docker, Linux, PySpark, SLURM, Snowflake
Robotic Platforms: Franka FR3 and Panda, Unitree (G1 and Go2), UR (10e, 5e), ORCA hand, Tactile Sensors